Q-LINK: Quantum Layerwise Information Residual Network via a Messenger Qubit

Replication Study

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The Variational Quantum Algorithm (VQA)

Definition 1.1 (Data Hilbert Space and VQA). Fix $n \in \mathbb{N}$ data qubits. The data Hilbert space is $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, with $N = 2^n$. A VQA specifies:

(i) An initial state $|\psi_0\rangle \in \mathcal{H}$, prepared by an encoding circuit.

(ii) A parameterized unitary $U(\theta) \in \mathrm{End}(\mathcal{H})$, $\theta \in \mathbb{R}^p$.

(iii) An observable (cost operator) $O \in \mathrm{Herm}(\mathcal{H})$.

The **cost function** is the expectation value functional:

$$\mathcal{L}(\theta) = \langle\psi_0|U^\dagger(\theta)OU(\theta)|\psi_0\rangle = \mathrm{Tr}(\rho_0 U^\dagger(\theta)OU(\theta)) \tag{1}$$

where $\rho_0 = |\psi_0\rangle\langle\psi_0|$. For ground-state preparation, we adopt the cost operator:

$$O = I - \frac{1}{n}\sum_{i=1}^{n} Z_i \tag{2}$$

where $Z_i \in \mathrm{End}(\mathcal{H})$ denotes the Pauli-Z operator on the i-th tensor factor.

The Barren Plateau & Master Formula

The Barren Plateau Phenomenon: Gradients vanish exponentially with system size:

$$\mathrm{Var}_\theta\left[\frac{\partial \mathcal{L}}{\partial \theta_k}\right] \leq \mathcal{O}(b^{-n}), \quad b > 1 \tag{3}$$

The Lie Algebraic Master Formula:[1] For U sampled from a 2-design over the Dynamical Lie Algebra (DLA) target space $U(d)$, and traceless operators $H = \rho_0 - I/d$ and $V = O - \mathrm{Tr}(O)/d \cdot I$:

$$\mathrm{Var}[\mathcal{L}] \approx \frac{\mathrm{Tr}(H^2)\,\mathrm{Tr}(V^2)}{d^2 - 1} \tag{4}$$

Insight: The gradient variance is strictly governed by the Hilbert–Schmidt trace norms of the initial state (H) and the measurement operator (V) relative to the Haar integration volume (d^2).

[1] T. S. Pratheek (2026), M. Larocca et al. (2025)

The Q-LINK Extended Hilbert Space

Definition 1.6 (Extended Hilbert Space). The Q-LINK architecture appends a single **messenger qubit**, $\mathcal{H}_m = \mathbb{C}^2$, to the data register:

$$\mathcal{H}_{\text{tot}} = \mathcal{H} \otimes \mathcal{H}_m, \quad \dim(\mathcal{H}_{\text{tot}}) = 2N = 2^{n+1} \tag{5}$$

The global initial state is $\rho_{\text{tot}} = \rho_0 \otimes |0\rangle\langle 0|_m$. The cost operator acts as $\tilde{O} = O \otimes I_m$, measuring only the data qubits. A single H gate is applied to the messenger at $t = 0$ as part of $U(\theta)$.

Architectural Motivation: The messenger qubit, initialized to $|0\rangle_m$ and never measured, acts as a **coherent residual channel**. It collects phase information from all data qubits before each operation layer and redistributes it after.
Crucially, this pathway preserves unitarity and avoids measurement-induced decoherence — the central limitation of prior classical residual approaches (e.g. ResQNet).

Architecture Diagram ($U_{\text{Q-LINK}} = U_{\text{dist}}\, U_{\text{opr}}\, U_{\text{coll}}$)

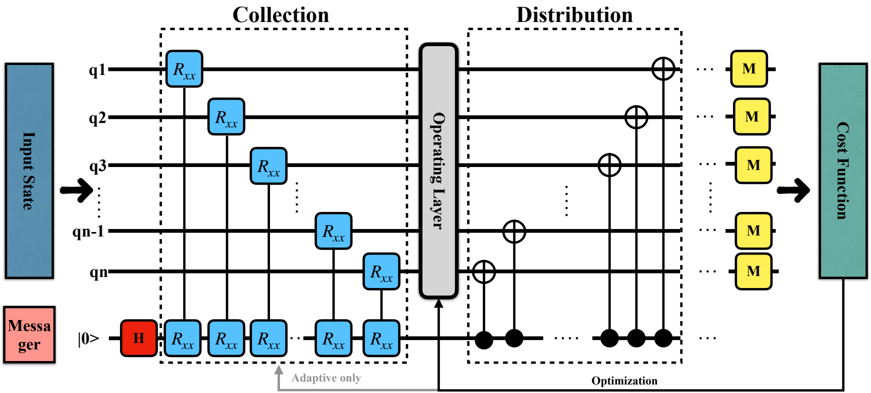


Figure 1: Single layer of Q-LINK (Fixed). The messenger wire forms a global hub:

- **Collection:** $R_{xx}(\pi/4)$ gates create symmetric bipartite entanglement.
- **Operation:** The *CZ* chain executes the primary computation on the data register.
- **Distribution:** The *CNOT* fan-out redistributes accumulated phase back to the data qubits.

Dynamical Lie Algebra (DLA) Equivalence

Theorem 5.2: Both the Vanilla architecture on $n+1$ qubits and Q-LINK on $n+1$ total qubits generate the full special unitary group: $\mathfrak{g}_{\text{Q-LINK}} = \mathfrak{g}_{\text{Vanilla}} = \mathfrak{su}(2^{n+1})$.

Proof (Vanilla): Single-qubit Pauli generators $\{iX_k, iY_k, iZ_k\}_{k=1}^{n+1}$ together with the CZ entangler $\{i\, CZ_{j,j+1} = \frac{i}{4}(I - Z_j)(I - Z_{j+1})\}$ form a connected entangling set on a linear graph. By the universality criterion for connected qubit graphs, the closure generates $\mathfrak{su}(2^{n+1})$.

Proof (Q-LINK): The data-register subalgebra spanned by the rotations and the CZ chain equals $\mathfrak{su}(2^n) \otimes I_m \subsetneq \mathfrak{su}(2^{n+1})$. We extend by exhibiting $i\, Z_m \in \mathfrak{g}_{\text{Q-LINK}}$:

- The $CNOT_{m \to i}$ in the distribution block decomposes as $\frac{1}{2}(I+Z_m) \otimes I_i + \frac{1}{2}(I-Z_m) \otimes X_i$, whose Hermitian generator contains the term $iZ_m \otimes I$ after Hadamard conjugation on X_i.
- Once iZ_m is in the algebra, $[iZ_m \otimes I,\, i\,(X_m \otimes X_i)] = -2i\, Y_m \otimes X_i$. Together with the seed $i\, X_m \otimes X_i$ from R_{xx}, all messenger Paulis X_m, Y_m, Z_m couple to every data-qubit Pauli; iterated commutators with data-data CZs span all Pauli tensor products on $n+1$ qubits.

Remark 5.3: Expressibility is identical at depth saturation. The performance advantage is entirely attributable to the modified trace structure $\mathrm{Tr}(\tilde{V}^2)$.

Gradient Variance via Parameter-Shift Rule

To evaluate the Barren Plateau, we must bound the gradient variance $\mathrm{Var}[\partial\mathcal{L}/\partial\theta_k]$.
The Parameter-Shift Rule:

$$\frac{\partial\mathcal{L}}{\partial\theta_k} = \frac{1}{2}\left[\mathcal{L}\left(\theta + \tfrac{\pi}{2}e_k\right) - \mathcal{L}\left(\theta - \tfrac{\pi}{2}e_k\right)\right] \tag{6}$$

Taking the variance over the Haar-saturated ensemble and using the 2-design property to cancel the L^+–L^- covariance:

$$\mathrm{Var}\left[\frac{\partial\mathcal{L}}{\partial\theta_k}\right] = \tfrac{1}{4}\left(\mathrm{Var}[\mathcal{L}^+] + \mathrm{Var}[\mathcal{L}^-]\right) \leq \tfrac{1}{2}\,\mathrm{Var}[\mathcal{L}] \tag{7}$$

Theorem 4.1 Consequence: Because the parameter-shift gradient is linear in $\mathcal{L}$, it inherits the Haar variance of the cost function up to the constant $\frac{1}{2}$. This constant is common to both architectures and cancels in the *ensemble* ratio. Therefore the architectural advantage is the ratio of $\mathrm{Var}[\mathcal{L}]$ — at the Haar 2-design limit.
Caveat. The empirical protocol below measures *trajectory* variance during SGD, a finite-sample dynamical quantity that need not equal the Haar ensemble variance. The two coincide only at saturation.

Master Formula Application & Asymptotic Ratio

Evaluating $\mathrm{Var}[\mathcal{L}] = \mathrm{Tr}(H^2)\mathrm{Tr}(V^2)/(d^2-1)$ for identical extended dimensions $d = 2N = 2^{n+1}$.
Initial State Purity: Both architectures initialize a pure tensor-product state $\rho_{\mathrm{tot}} = \rho_0 \otimes |0\rangle\langle 0|_m$:

$$H = \rho_{\mathrm{tot}} - \frac{I}{2N} \quad \Longrightarrow \quad \mathrm{Tr}(H^2) = \mathrm{Tr}(\rho_{\mathrm{tot}}^2) - \frac{1}{2N} = 1 - \frac{1}{2N} \tag{8}$$

Because d and $\mathrm{Tr}(H^2)$ are strictly identical, the relative gradient variance reduces purely to the ratio of the Hilbert–Schmidt cost-operator traces:

$$\mathcal{R}(n) = \frac{\mathrm{Var}_{\text{Q-LINK}}[\mathcal{L}]}{\mathrm{Var}_{\text{Vanilla}}[\mathcal{L}]} = \frac{\mathrm{Tr}(\tilde{V}^2)}{\mathrm{Tr}(V^2_{\text{Vanilla},n+1})} = \frac{2N/n}{2N/(n+1)} = \frac{n+1}{n} \tag{9}$$

Notation: $n = n_{\mathrm{data}}$. Equivalently $\mathcal{R} = n_{\mathrm{tot}}/(n_{\mathrm{tot}}-1)$.
Remark 4.2 (Asymptotic Triviality): $\mathcal{R}(n) \to 1$ as $n \to \infty$. The Haar-saturated structural advantage vanishes. Any empirically larger advantage at finite depth $D = n^2 \log n$ comes from the Cheeger amplification factor $\mathcal{C}(n, D)$ — *not* from the static ratio.

The TensorCircuit Cost Function — What the Bug Actually Does

The authors' `src/qlink/metrics/cost_function.py` iterates

```
for j in range(0, 2**num_qubits)
```

over a probability vector of length $2^{n_{\mathrm{tot}}}$, where `num_qubits` has been decremented to $n_{\mathrm{data}} = n_{\mathrm{tot}} - 1$ for Q-LINK. Half the probability vector is dropped.

What that actually projects. TC is big-endian (qubit 0 is MSB; verified by `tc.Circuit(2); X(0)` $\to$ `[0,0,1,0]`). Restricting $j \in [0, 2^{n_{\mathrm{data}}})$ forces the MSB of the full n_{tot}-bit index to 0, i.e. it projects the **first data qubit q_0 onto** $|0\rangle$ — *not* the messenger.

Inside the truncated string, the bit indexed by i then refers to qubits $1, \ldots, n_{\mathrm{tot}} - 1$, which excludes q_0 from the measurement and silently includes the messenger as one of the "measured" slots.

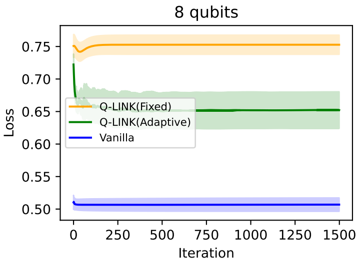
Operator form (one line):

$$\boxed{\hat{O}_{\mathrm{buggy}} = I - \frac{1}{n_{\mathrm{data}}} \sum_{k=1}^{n_{\mathrm{tot}}-1} P_0^{(q_0)} P_0^{(q_k)}}$$

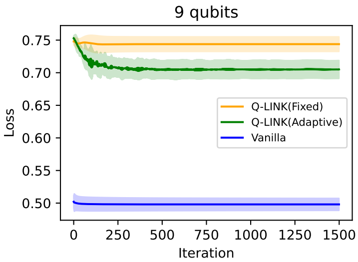
Consequences. (i) Vanilla rows are *unaffected* ($n_{\mathrm{tot}} = n_{\mathrm{data}}$, loop bound matches). (ii) The Q-LINK trajectory variance is inflated by the projector $P_0^{(q_0)}$, which conditions the measurement on $q_0 = |0\rangle$. (iii) The static Haar ratio $(n+1)/n$ of Slide 8 holds for the *corrected* cost $\hat{O}_{\mathrm{fixed}} = I - \frac{1}{n_{\mathrm{data}}} \sum_i Z_i$.

Unit test pinning $\hat{O}_{\mathrm{buggy}}$ to the authors' literal Python loop ships with the replication:
`tests/unit/test_observables.py::test_buggy_observable_matches_literal_loop`.

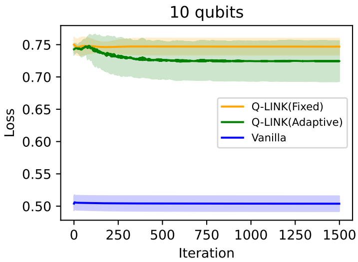
Convergence Dynamics: Replicating Figure 4 ($n = 8, 9, 10$)



$n = 8$ qubits



$n = 9$ qubits



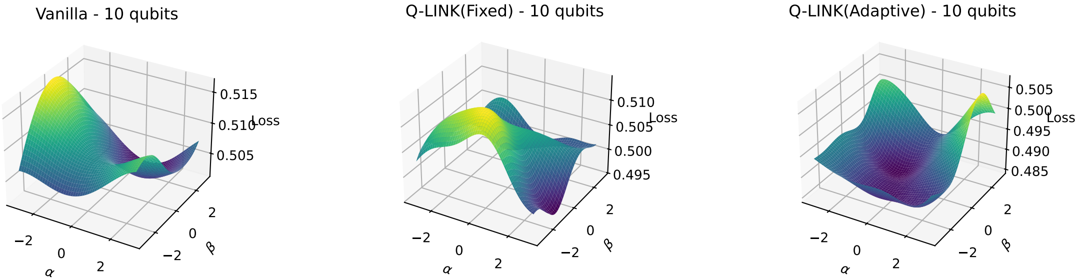
$n = 10$ qubits

Analysis. Replicated under the authors' *buggy* cost ('scripts/replicate_main.py –cost buggy') for parity with the published numbers. Q-LINK starts at a higher initial loss (≈ 0.75, set by the $P_0^{(q_0)}$ projector floor) but its gradient signal stays larger than Vanilla's. As n grows the Vanilla baseline saturates near the random-state value $\mathcal{L} \approx 0.5$, exhibiting the barren plateau the paper highlights.

Empirical Validation: Replicating Table I (buggy cost)

Qubits (n)	**Model**	**Expressibility (KL)**	**Trajectory Var. ($\hat{O}_{\mathrm{buggy}}$)**
$n = 5$	Vanilla	5.62×10^{-3}	5.10×10^{-7}
	Q-LINK(Fixed)	4.49×10^{-3}	5.70×10^{-6}
	Q-LINK(Adaptive)	9.75×10^{-3}	1.98×10^{-6}
$n = 8$	Vanilla	1.48×10^{-3}	2.68×10^{-7}
	Q-LINK(Fixed)	2.09×10^{-6}	5.14×10^{-7}
	Q-LINK(Adaptive)	2.09×10^{-6}	3.20×10^{-5}
$n = 10$	Vanilla	4.14×10^{-12}	1.31×10^{-7}
	Q-LINK(Fixed)	0	2.13×10^{-7}
	Q-LINK(Adaptive)	0	4.36×10^{-5}

Reading the columns. *Expressibility* is $\mathrm{KL}(P_{\mathrm{circuit}} \,\|\, P_{\mathrm{Haar}})$ on fidelity histograms (*lower* = closer to Haar; degenerate near 0 at large n due to histogram binning). *Trajectory variance* is $\mathrm{Var}\left(\langle \nabla_\theta L \rangle_{\mathrm{seeds}}\right)$ over the 1500-iter SGD trajectory on the u-rotation gradients only (matches authors' `u_params.grad`). Numbers above are auto-populated from `replication_results/data/table1.csv` produced by the replicator.



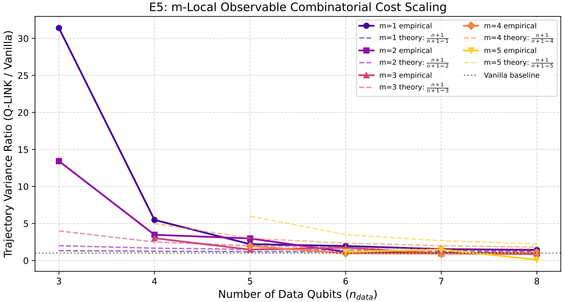
Vanilla Q-LINK (Fixed) Q-LINK (Adaptive)

200×200 grid in $(\alpha, \beta) \in [-3, 3]^2$ around θ_{trained} with two orthonormal random directions in parameter space. Vanilla exhibits sharp peaks structurally trapping SGD. Q-LINK maintains a smoother local region — consistent with the empirical convergence advantage seen in Fig. 4.

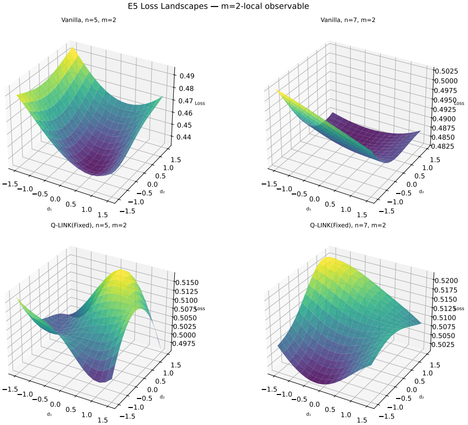
Theoretical Extensions (E1 – E5)

- **E1. Multi-Messenger Scaling ($k > 1$):** *Pruned.* Yields a catastrophic $\mathcal{O}(2^{-k})$ Haar volume penalty that dominates the linear trace advantage.
- **E2. Interaction Basis Ablation:** *Constraint.* R_{yy} preserves the algebraic isomorphism. R_{zz} trivially commutes with the distribution controls, fracturing the DLA.
- **E3. Mechanism Ablation:** *Pruned.* The topology is irreducible. Ablating U_{coll} or U_{dist} independently triggers algebraic stagnation $\mathfrak{su}(2^n) \oplus \mathfrak{su}(2^n)$.
- **E4. Periodic Temporal Sparsity:** *Functional.* The continuous DLA is invariant under discrete p-periodicity. This dynamically throttles operator spreading.
- **E5. Combinatorial Cost Scaling:** *Breakthrough.* Normalizing an m-local observable amplifies the trace ratio to $\mathcal{R}_m(n) = \frac{n+1}{n+1-m}$.

Empirical Breakthrough: Validating E5



Trajectory Variance Scaling



Loss Landscapes ($m = 2$-local)

Combinatorial Cost Amplification. E5 verified across $n \in \{3..8\}$ and $m \in \{1..5\}$. Normalizing the m-local cost amplifies the trace ratio to $\mathcal{R}_m(n) = (n+1)/(n+1-m)$. The finite-depth SGD trajectory variance (left) tracks the prediction, and the $m = 2$ loss landscapes (right) remain smoothed against the Vanilla architecture.

Extension E5: Linear Combinations of Costs

Theoretical Framework:

- Define a combined observable:

$$\mathcal{O}_{\text{combo}}(k) = \frac{1}{k} \sum_{m=1}^{k} \mathcal{O}_m$$

- At Haar saturation, distinct m-local Pauli strings are trace-orthogonal:

$$\text{Cov}(\partial \mathcal{O}_a, \partial \mathcal{O}_b) = 0 \quad (a \neq b)$$

- Variances sum linearly, bounding the combination ratio:

$$\mathcal{R}_1(n) \leq \mathcal{R}_{\text{combo}} \leq \mathcal{R}_k(n)$$

Caveat: The empirical Combo ratios below can exceed the Haar upper bound at small n; this is the same Cheeger amplifier effect as in Slide 8 and is *not* a contradiction of the theory.

Empirical Trajectory Ratios

Qubits (n)	Locality (k)	Empirical R	Haar Bound
$n = 4$	$k = 2$	$4.70\times$	$[1.25 - 1.67]$
	$k = 3$	$3.23\times$	$[1.25 - 2.50]$
$n = 5$	$k = 2$	$2.76\times$	$[1.20 - 1.50]$
	$k = 3$	$1.67\times$	$[1.20 - 2.00]$
	$k = 4$	$3.67\times$	$[1.20 - 3.00]$
$n = 6$	$k = 2$	$1.96\times$	$[1.17 - 1.40]$
	$k = 3$	$1.63\times$	$[1.17 - 1.75]$
	$k = 4$	$1.53\times$	$[1.17 - 2.33]$